Electricity Pricing Reform in California

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My proposal

1. Take MPC and MSC values from literature (Borenstein and Bushnell)
2. Set new $p = \text{MSC}$
3. Find fixed fee under profit-neutral constraint

Table 1: Model Parameters and Policy Design

Parameter	Value
Price Elasticity of Demand	-0.415
New Marginal Price (MSC)	$0.07
New Fixed Fee (non-CARE)	$229.57
Frozen Fixed Fee (CARE)	$55.00
Number of Observations (PGE customers)	1,414,978

Figure 1: New pricing schedule

My figure

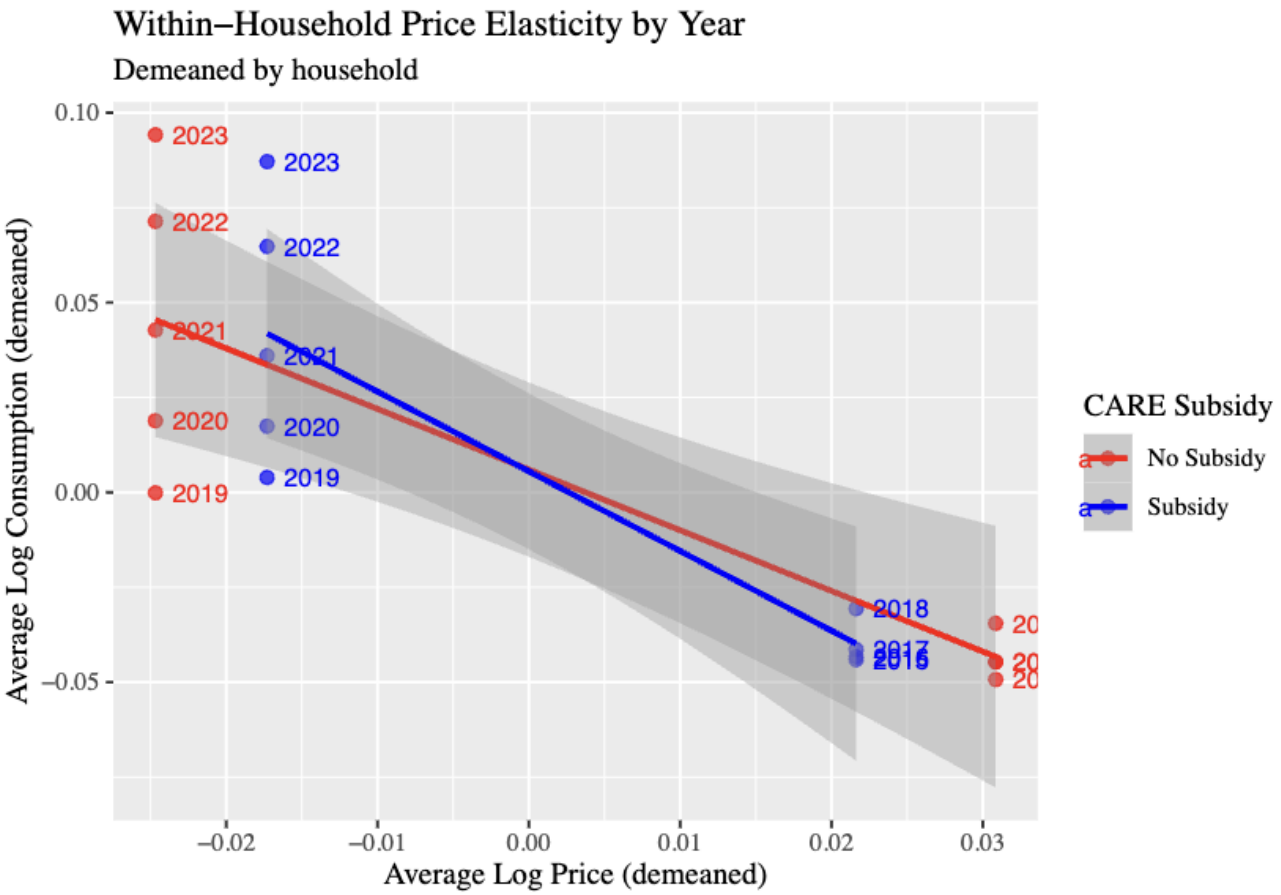

Figure 2: CARE regression graph

Comparison to baseline

Table 3: Policy Impact Analysis: Marginal Cost Pricing (PGE)

Metric	Value
Profits	
Baseline Profit	$329,695,607.01
Post-Policy Profit	$329,695,607.01
Change in Profit	$0.00
Consumption	
Baseline Consumption (kWh)	$1,425,212,199.27
Post-Policy Consumption (kWh)	$1,837,025,505.08
Change in Consumption (kWh)	$411,813,305.81
Welfare (Unweighted)	
Change in Consumer Surplus	$43,909,240.65
Change in Environmental Damages	$12,256,552.33
Change in Total Surplus (Unweighted)	$56,165,792.99
Welfare (Income-Weighted)	
Change in Consumer Surplus (Income-Weighted)	$13,793,603.22
Change in Total Surplus (Income-Weighted)	$26,050,155.55
Welfare (CARE-Weighted)	
Change in Consumer Surplus (CARE-Weighted)	$53,420,817.42
Change in Total Surplus (CARE-Weighted)	$65,677,369.75

Figure 3: Changes in surplus